

4.6.4

Taraka Abhinav - EE25BTECH11016

Question:

Find the equation of a line passing through the point $(2, 3, 2)$ and parallel to the line $\mathbf{r} = (-2\hat{i} + 3\hat{j}) + \lambda(2\hat{i} - 3\hat{j} + 6\hat{k})$. Also, find the distance between these two lines.

Solution:

1. Equation of the Line

The equation is $\mathbf{r} = \mathbf{a}_2 + \mu\mathbf{b}$.

$$\mathbf{r} = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix} \quad (0.1)$$

2. Distance Between Lines

The distance is given by the formula:

$$d = \frac{\|\mathbf{b} \times (\mathbf{a}_2 - \mathbf{a}_1)\|}{\|\mathbf{b}\|} \quad (0.2)$$

First, we find the required vectors algebraically. Let $\Delta\mathbf{a} = \mathbf{a}_2 - \mathbf{a}_1$. Let the cross product be $\mathbf{v} = \mathbf{b} \times \Delta\mathbf{a}$. The formula is $d = \frac{\|\mathbf{v}\|}{\|\mathbf{b}\|}$.

Now, we substitute the numerical values:

$$\mathbf{a}_1 = \begin{pmatrix} -2 \\ 3 \\ 0 \end{pmatrix}, \quad \mathbf{a}_2 = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix} \quad (0.3)$$

$$\Delta\mathbf{a} = \begin{pmatrix} 4 \\ 0 \\ 2 \end{pmatrix} \quad (0.4)$$

$$\mathbf{v} = \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix} \times \begin{pmatrix} 4 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} -6 \\ 20 \\ 12 \end{pmatrix} \quad (0.5)$$

$$\|\mathbf{v}\| = \sqrt{(-6)^2 + 20^2 + 12^2} = \sqrt{580} = 2\sqrt{145} \quad (0.6)$$

$$\|\mathbf{b}\| = \sqrt{2^2 + (-3)^2 + 6^2} = \sqrt{49} = 7 \quad (0.7)$$

The distance is:

$$d = \frac{2\sqrt{145}}{7} \quad (0.8)$$

